In [2]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

In [3]:
```
df = pd.read_csv(r"C:\Users\User\Desktop\dv.csv")
df
```

Out[3]:

| | name | planet_status | mass | mass_error_min | mass_error_max | mass_sini | mass_sini |
|---|---|---|---|---|---|---|---|
| **0** | 109 Psc b | Confirmed | 5.743 | 0.28900 | 1.01100 | 6.38300 | |
| **1** | 112 Psc b | Confirmed | NaN | 0.00500 | 0.00400 | 0.03300 | |
| **2** | 112 Psc c | Confirmed | 9.866 | 1.78100 | 3.19000 | NaN | |
| **3** | 11 UMi b | Confirmed | NaN | 1.10000 | 1.10000 | 11.08730 | |
| **4** | 14 And Ab | Confirmed | NaN | 0.23000 | 0.23000 | 4.68400 | |
| **...** | ... | ... | ... | ... | ... | ... | |
| **5981** | YBP 401 b | Confirmed | NaN | 0.05000 | 0.05000 | 0.46000 | |
| **5982** | YSES 2 b | Confirmed | 6.300 | 0.90000 | 1.60000 | NaN | |
| **5983** | YZ Cet b | Confirmed | NaN | 0.00028 | 0.00028 | 0.00220 | |
| **5984** | YZ Cet c | Confirmed | NaN | 0.00035 | 0.00035 | 0.00359 | |
| **5985** | YZ Cet d | Confirmed | NaN | 0.00038 | 0.00038 | 0.00343 | |

5986 rows × 10 columns

In [4]:
```
print("Shape:", df.shape)
```

```
Shape: (5986, 10)
```

In [5]:
```
df.fillna("Nil")
```

Out[5]:

| | name | planet_status | mass | mass_error_min | mass_error_max | mass_sini | mass_sini |
|---|---|---|---|---|---|---|---|
| **0** | 109 Psc b | Confirmed | 5.743 | 0.289 | 1.011 | 6.383 | |
| **1** | 112 Psc b | Confirmed | Nil | 0.005 | 0.004 | 0.033 | |
| **2** | 112 Psc c | Confirmed | 9.866 | 1.781 | 3.19 | Nil | |
| **3** | 11 UMi b | Confirmed | Nil | 1.1 | 1.1 | 11.0873 | |
| **4** | 14 And Ab | Confirmed | Nil | 0.23 | 0.23 | 4.684 | |
| **...** | ... | ... | ... | ... | ... | ... | |
| **5981** | YBP 401 b | Confirmed | Nil | 0.05 | 0.05 | 0.46 | |
| **5982** | YSES 2 b | Confirmed | 6.3 | 0.9 | 1.6 | Nil | |
| **5983** | YZ Cet b | Confirmed | Nil | 0.00028 | 0.00028 | 0.0022 | |
| **5984** | YZ Cet c | Confirmed | Nil | 0.00035 | 0.00035 | 0.00359 | |
| **5985** | YZ Cet d | Confirmed | Nil | 0.00038 | 0.00038 | 0.00343 | |

5986 rows × 10 columns

In [6]:

```
print("\nColumns and Data Types:")
print(df.info())
```

```
Columns and Data Types:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5986 entries, 0 to 5985
Data columns (total 10 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   name                5986 non-null   object
 1   planet_status       5986 non-null   object
 2   mass                2079 non-null   float64
 3   mass_error_min      2817 non-null   float64
 4   mass_error_max      2817 non-null   float64
 5   mass_sini           1202 non-null   float64
 6   mass_sini_error_min 1039 non-null   float64
 7   mass_sini_error_max 1039 non-null   float64
 8   radius              4568 non-null   float64
 9   radius_error_min    4093 non-null   float64
dtypes: float64(8), object(2)
memory usage: 467.8+ KB
None
```

In [7]:
```
print("\nFirst 5 rows:")
print(df.head())
```

```
First 5 rows:
        name planet_status   mass  mass_error_min  mass_error_max  mass_sini  \
0  109 Psc b     Confirmed  5.743           0.289           1.011     6.3830   
1  112 Psc b     Confirmed    NaN           0.005           0.004     0.0330   
2  112 Psc c     Confirmed  9.866           1.781           3.190        NaN   
3   11 UMi b     Confirmed    NaN           1.100           1.100    11.0873   
4  14 And Ab     Confirmed    NaN           0.230           0.230     4.6840   

   mass_sini_error_min  mass_sini_error_max  radius  radius_error_min  
0                0.078                0.078   1.152               NaN  
1                0.005                0.004     NaN               NaN  
2                  NaN                  NaN     NaN               NaN  
3                1.100                1.100     NaN               NaN  
4                0.230                0.230     NaN               NaN  
```

In [8]:
```
print("Planet Status counts:")
print(df['planet_status'].value_counts())
print("\nUnique names or duplicates:")
print("Duplicate rows:", df.duplicated().sum())
print("Duplicate names:", df['name'].duplicated().sum())
print("\nMissing values summary:")
print(df.isnull().sum())
```

```
Planet Status counts:
planet_status
Confirmed    5986
Name: count, dtype: int64

Unique names or duplicates:
Duplicate rows: 0
Duplicate names: 0

Missing values summary:
name                      0
planet_status             0
mass                   3907
mass_error_min         3169
mass_error_max         3169
mass_sini              4784
mass_sini_error_min    4947
mass_sini_error_max    4947
radius                 1418
radius_error_min       1893
dtype: int64
```

In [9]:
```
print("Numeric Descriptive Statistics:")
print(df.describe())
```

```
Numeric Descriptive Statistics:
              mass  mass_error_min  mass_error_max    mass_sini  \
count  2079.000000     2817.000000       2817.0000  1202.000000
mean      1.492538        0.324366             inf     1.983529
std       2.602286        1.351929             NaN     3.199717
min       0.000190        0.000000        -17.8000     0.000470
25%       0.029655        0.006290          0.0057     0.045000
50%       0.370000        0.041000          0.0400     0.731500
75%       1.466500        0.170000          0.1700     2.400000
max      12.934000       53.500000             inf    50.000000

       mass_sini_error_min  mass_sini_error_max       radius  radius_error_min
count          1039.000000          1039.000000  4568.000000       4093.000000
mean              0.214687             0.240453     0.401515          0.048986
std               0.636474             0.945468     0.430601          0.754220
min               0.000000             0.000000     0.025430          0.000000
25%               0.005940             0.005300     0.145000          0.010000
50%               0.049300             0.050000     0.219000          0.020000
75%               0.170000             0.170000     0.384000          0.042000
max              13.000000            20.900000     6.900000         48.000000
```

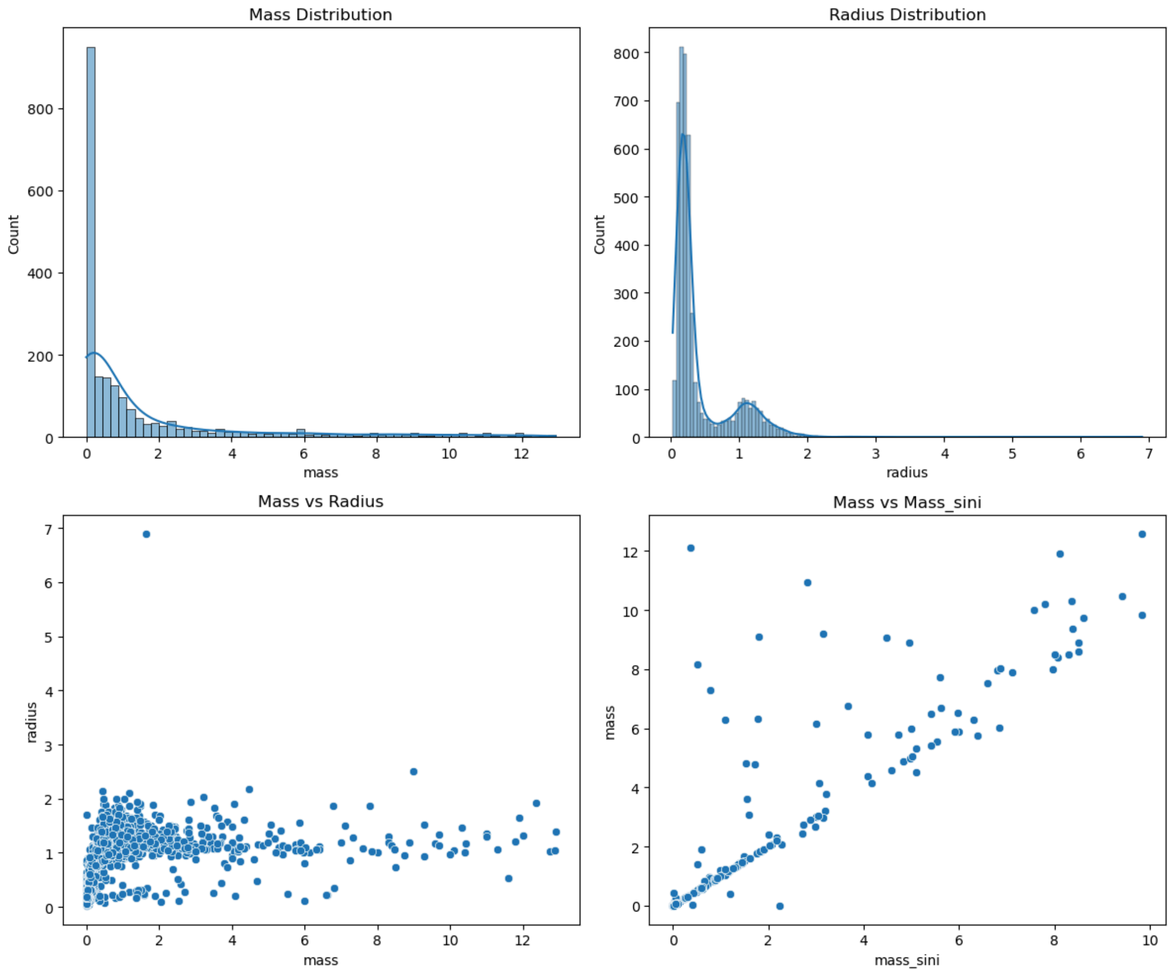
In [10]:

```
print("Infinite values in mass_error_max:", np.isinf(df['mass_error_max']).sum()
print("Negative values in mass_error_max:", (df['mass_error_max'] < 0).sum())
print("Rows with negative mass_error_max:")
print(df[df['mass_error_max'] < 0])
print("\nRows with inf mass_error_max:")
print(df[np.isinf(df['mass_error_max'])])
```

```
Infinite values in mass_error_max: 9
Negative values in mass_error_max: 1
Rows with negative mass_error_max:
           name planet_status  mass  mass_error_min  mass_error_max  \
803  HD 19615 b     Confirmed   8.6            53.5           -17.8

     mass_sini  mass_sini_error_min  mass_sini_error_max  radius  \
803        8.5                  0.7                  0.7     NaN

     radius_error_min
803               NaN

Rows with inf mass_error_max:
             name planet_status      mass  mass_error_min  mass_error_max  \
92       CD Cet b     Confirmed   0.01244         0.00136             inf
768   HD 181720 b     Confirmed  12.12000         5.88000             inf
1093   HD 69830 b     Confirmed   0.14270         0.09280             inf
1094   HD 69830 c     Confirmed   0.16490         0.10720             inf
1095   HD 69830 d     Confirmed   0.25290         0.16440             inf
5238   TOI-1416 c     Confirmed   0.06800         0.00880             inf
5529    TOI-431 c     Confirmed   0.00892         0.00107             inf
5701    TOI-784 c     Confirmed   0.06878         0.02912             inf
5749   ups And Ab     Confirmed   0.62000         0.09000             inf

      mass_sini  mass_sini_error_min  mass_sini_error_max  radius  \
92      0.01243              0.00135              0.00135     NaN
768     0.37000                  NaN                  NaN     NaN
1093    0.03210                  NaN                  NaN     NaN
1094    0.03710                  NaN                  NaN     NaN
1095    0.05690                  NaN                  NaN     NaN
5238    0.06800              0.00880              0.00980     NaN
5529    0.00890              0.00107              0.00129     NaN
5701    0.06875              0.02910              0.02910     NaN
5749    0.62000              0.09000              0.09000     NaN

      radius_error_min
92                 NaN
768                NaN
1093               NaN
1094               NaN
1095               NaN
5238               NaN
5529               NaN
5701               NaN
5749               NaN
```

In [11]:
```python
fig, axes = plt.subplots(2, 2, figsize=(12, 10))
sns.histplot(df['mass'].dropna(), ax=axes[0,0], kde=True).set_title('Mass Distri
sns.histplot(df['radius'].dropna(), ax=axes[0,1], kde=True).set_title('Radius Di
sns.scatterplot(data=df, x='mass', y='radius', ax=axes[1,0]).set_title('Mass vs
sns.scatterplot(data=df, x='mass_sini', y='mass', ax=axes[1,1]).set_title('Mass
plt.tight_layout()
plt.savefig('distributions.png')
print("Plots saved.")
```

```
Plots saved.
```




In [12]:
```
corr = df[['mass', 'mass_sini', 'radius']].corr()
print("Correlation Matrix:")
print(corr)
```

```
Correlation Matrix:
                mass  mass_sini    radius
mass        1.000000   0.875983  0.417286
mass_sini   0.875983   1.000000  0.130681
radius      0.417286   0.130681  1.000000
```

In [ ]: